



Figure 1: Evolution of $\mathbf{p}^{(m)}$ in the *Fully Synthetic* setting for vocabulary size $s = 3$, context length $\ell = 4$, total contexts $c = s^\ell = 81$ and sample size $n = 1000$. The initial distribution $\mathbf{p}^{(0)}$ is some random distribution over tokens. The trained conditional distributions converge towards Dirac measures over generations illustrating *total collapse* in Definition 1.

2.2 Recursive Training

In this section, we introduce the notations for recursive training. At generation $m \geq 1$, suppose that we have samples $\{\mathbf{y}_l^{(t)}\}_l$ generated by some past models $\mathbf{p}^{(t)}$ for $t \in \{0, \dots, m-1\}$ respectively. As such, similarly to (2), the model at generation m expresses as

$$\mathbf{p}^{(m)} := \frac{1}{n^{(m)}} \sum_{t=0}^{m-1} \sum_{l=1}^{n_t^{(m)}} \mathbf{y}_l^{(t)}. \quad (5)$$

In other words, $\mathbf{p}^{(m)}$ is obtained by training the model on a mixture of real and synthetic data generated from previously trained models. Here, $n_t^{(m)}$ stands for the number of samples used to train the m -th generation model that are generated by model $\mathbf{p}^{(t)}$, and $n^{(m)} = n_0^{(m)} + \dots + n_{m-1}^{(m)}$ is the total number of training samples used to train $\mathbf{p}^{(m)}$. Note that we do not generate samples from $\mathbf{p} = \mathbf{p}^{(0)}$ but rather we are given a corpus $\{\mathbf{y}_l^{(0)}\}_l$ which represent original data.

Note that by definition all the models $\mathbf{p}^{(m)}$ are unbiased, i.e. $\mathbb{E}\mathbf{p}^{(m)} = \mathbf{p}$, which means that they recover the original distribution in case of infinite sample sizes. However, in practical settings the sample sizes are finite, and therefore $\mathbf{p}^{(m)}$ may deviate from $\mathbf{p}$ since its variance can be large for small values of $n_t^{(m)}$. In the remainder of the paper, we present analytical results to rigorously quantify the impact of sample sizes $n_t^{(m)}$ on the variance of $\mathbf{p}^{(m)}$ and how they affect the learned distribution to eventually cause *model collapse* (Shumailov et al., 2023), a degenerative process affecting future generation models by either losing information about the tails of the initial distribution or inducing distribution shifts over generations.

Specifically, we aim to quantify the rate of such deterioration, and to this end, we introduce a stricter version of model collapse that we call *total collapse* and define as follows.

Definition 1 (Total Collapse). *We say that total collapse occurs in the recursive training process if $\mathbf{p}^{(m)}$ converges to some Dirac mass δ_i for some token $i \in [s]$ as m grows.*

Total collapse refers to the case where, under recursive training, the trained model $\mathbf{p}^{(m)}$ completely loses information about the original distribution $\mathbf{p}$ over generations, leading to poor linguistic diversity.¹ This phenomenon is illustrated in Figure 1 where we see convergence towards the vertices of the probability simplex, i.e. Dirac measures, over generations. With this definition of total collapse, we provide a quantitative analysis of two cases of recursive training:

¹Note that here we define total collapse with respect to a single pre-fixed context. This can be generalized to the event where total collapse occurs for all contexts. However, the theoretical analysis in this case would require more refined treatment of several statistical quantities to obtain uniform bounds over contexts. We leave this for future work.

- *Fully Synthetic*: Training with synthetic data from the last model. Each generation $\mathbf{p}^{(m)}$ is trained only on data generated by the previous model $\mathbf{p}^{(m-1)}$. More precisely, for some fixed $n \in \mathbb{N}$ we let $n_t^{(m)} = n \cdot \mathbb{1}\{t = m - 1\}$ for all $m \geq 1$ and $0 \leq t < m$.
- *Partially Synthetic*: Training with a mixture of real and synthetic data from the last model. Each generation $\mathbf{p}^{(m)}$ is trained on a mixture of real data and synthetic data generated by the previous model $\mathbf{p}^{(m-1)}$. More precisely, for some fixed $n \in \mathbb{N}$ we let $n_0^{(m)} = N$, $n_{m-1}^{(m)} = n$ and $n_t^{(m)} = 0$ for all $m \geq 2$ and $0 < t < m$, and $n_0^{(1)} = N$.

Fully Synthetic corresponds to the theoretical setting considered in (Shumailov et al., 2023). However, in that paper, only Gaussian distribution was analyzed instead of discrete distributions over all possible tokens as language models entail. We point out that this setting is unlikely to happen in real-world applications but serves as the worst-case scenario.

On the other hand, the *Partially Synthetic* setting is more realistic for future generation models as it corresponds to training on a mixture of real and synthetic data. We consider this setting to assess whether it is possible to avoid collapse by having a fraction of the original data in the training mixture. We answer this question positively in the next section. Moreover, we show through simulations in Section 4.1 that conclusions from our theoretical analysis on both *Fully Synthetic* and *Partially Synthetic* hold beyond these simple settings, such as training on a mixture of all generations or even using realistic transformer models.

3 Main Results

To investigate the model collapse phenomenon and the rate at which it occurs, we define the following statistical quantities that capture the randomness of $\mathbf{p}^{(m)}$:

$$\|\mathbf{p}^{(m)}\|_\infty := \max_{i \in [s]} p_i^{(m)}, \quad \sigma_m := \|\mathbf{p}^{(m)}\|_2^2 = \sum_{i=1}^s p_i^{(m)2} \quad \text{and} \quad S_m := \mathbb{E}[\sigma_m].$$

These quantities measure how far away $\mathbf{p}^{(m)}$ is from some Dirac mass (Total Collapse, Definition 1). Since the maximum value of all three quantities is 1, the closer they are to 1, the closer $\mathbf{p}^{(m)}$ is to some Dirac mass, and the less diverse $\mathbf{p}^{(m)}$ is as a language model. As such, $\|\mathbf{p}^{(m)}\|_\infty$ or σ_m being equal to 1 is equivalent to $\mathbf{p}^{(m)}$ being a Dirac mass which reflects total collapse. To quantify the distribution shift incurred by the aforementioned recursive training scenarios, we further consider the 1-norm² between two distributions $\mu, \nu \in \mathbb{R}^s$ defined by $\|\mu - \nu\|_1 := \sum_{i=1}^s |\mu(i) - \nu(i)|$.

We assume that the initial distribution $\mathbf{p}^{(0)}$ is *nontrivial*, specifically $S_0 < 1$ and $\|\mathbf{p}^{(0)}\|_\infty < 1$. Under this assumption, we provide results on the rate of total collapse and further quantify distribution shift under recursive training. All the proofs are presented in Appendix B.

3.1 *Fully Synthetic*: Training on synthetic data

We start by describing the recursive training process in which only synthetic data from the last generation model are used. This process can be viewed as a Markov chain on the set of probability measures $\Delta^{s-1} \cap \frac{1}{n} \mathbb{N}^s$ on $[s]$ with denominator n , where

$$\Delta^{s-1} := \{\mathbf{v} \in \mathbb{R}^s : v_1 + v_2 + \dots + v_s = 1, v_i \geq 0 \text{ for all } i\}$$

²We point out that the 1-norm is twice the total variation distance $\|\mu - \nu\|_{\text{TV}}$, which is another commonly used metric for probability distributions and was considered by Fu et al. (2024) for studying model collapse in the case of diffusion models.

is the probability simplex. Since the probability of reaching δ_i is positive provided $p_i > 0$, this random walk which starts at $\mathbf{p}^{(0)}$ has absorbing states $\{\delta_i : i \in [s] \text{ s.t. } p_i > 0\}$. As a result, the random walk converges to one of the absorbing states almost surely (Kemeny et al., 1969), which means that total collapse is bound to happen in the *Fully Synthetic* setting.

To characterize the rate at which total collapse occurs, let us denote by $T := \inf\{t \in \mathbb{N} : \|\mathbf{p}^{(t)}\|_\infty = 1\} \geq 1$ the random time at which the model first becomes a Dirac mass, and let $\rho_m := \mathbb{P}(\|\mathbf{p}^{(m)}\|_\infty = 1)$ denote the probability that the m -th generation has collapsed. Our first result, presented in Theorem 1, provides the rate of convergence via S_m , ρ_m and $\mathbb{E}[T]$.

Theorem 1 (Control on Total Collapse). *Consider the Fully Synthetic setting and let $\tilde{s} := |\text{supp}(\mathbf{p})|$ denote the support size of $\mathbf{p}$, namely $\tilde{s} := |\{i \in [s] : p_i > 0\}|$.*

1. *The expected sum of squared probabilities S_m is given by*

$$S_m = 1 - \left(1 - \frac{1}{n}\right)^m (1 - S_0). \quad (6)$$

2. *The probability ρ_m that total collapse has occurred by generation m satisfies*

$$1 - n(1 - S_0)(1 - 1/n)^m \leq \rho_m \leq 1 - \frac{1 - S_0}{1 - 1/\tilde{s}}(1 - 1/n)^m. \quad (7)$$

3. *The generation T at which total collapse happens satisfies*

$$1 + \frac{1 - S_0}{1 - 1/\tilde{s}}(n - 1) \leq \mathbb{E}[T] \leq 1 + (1 - S_0)n(n - 1). \quad (8)$$

In essence, Theorem 1 describes the behavior of $\mathbf{p}^{(m)}$ as a function of the model generation m , the sample size n , and the dispersion of the initial distribution S_0 . Specifically, we draw the following observations:

- Effect of the number of generations m : As m increases, S_m and ρ_m tend to 1 as per (6) and (7), making total collapse increasingly more likely. Note also that this dependence is exponential in m , making total collapse in this case relatively fast.
- Effect of “synthetic” sample size n : The smaller n is, the more likely $\mathbf{p}^{(m)}$ is to have collapsed for a given generation m as per the bounds on ρ_m in (7), and the faster total collapse is expected to happen as suggested by (8). The dependence in this case is polynomial in n .
- Effect of S_0 : Larger values of S_0 correspond to faster collapse as suggested by (8); namely, starting from an original distribution that is not diverse enough speeds up total collapse. The upper and lower bounds on $\mathbb{E}[T]$ are both linear in $(1 - S_0)$.

We point out that when the number of contexts c is large, the number of samples n per context would be fairly small, which suggests that total collapse, given a single context, is expected to happen fairly quickly. The bounds on $\mathbb{E}[T]$ state that the expected collapse time is at least of order n and at most of order n^2 , though experiments (see Figure 2) suggest that the upper bound is not sharp and should be close to $\mathcal{O}(n)$ instead. On another note, we characterize below in Proposition 1 the limiting distribution as m gets to infinity, which demonstrates the direction of total collapse.

Proposition 1. *In the Fully Synthetic case, we have $\mathbb{P}\left(\lim_{m \rightarrow \infty} \mathbf{p}^{(m)} = \delta_i\right) = p_i$ for all $i \in [s]$.*

Proposition 1 describes the limiting distribution when total collapse occurs in terms of the initial probabilities p_i over tokens. Specifically, the resulting Dirac mass δ_i is likely to be supported on some token i with high initial probability p_i . This formally supports the description of *early* and *late* model collapse in Shumailov et al. (2023): In the early phase of recursive training, the tails of the original distribution disappear because the probability